Registrar workflow state machine: map course schedule approvals from department draft to Academic Affairs and final publish

Computer
RBAC permission matrix: map view, edit, export, and approval rights across Registrar, Department Chair, and Dean roles
Computer
Show the rest of the state machine table and transition rules
Map out the role-based permission matrix for department schedulers
Show the conflict detection engine for instructor and room overlaps

A strong registrar workflow should move from department draft through chair/dean review, registrar validation, Academic Affairs exceptions when needed, and final publication, with rule-based checks for meeting patterns, room use, and overlaps. University scheduling policies commonly require standard meeting patterns, designated departmental submissions, chair/dean review, registrar conformity checks, and special approval for non-standard times or exceptional offerings. [1] [2] [3]

Approval state machine

The full schedule workflow should separate ordinary approvals from exception approvals, because non-standard meeting times, special room needs, and unusual offerings often require dean or Academic Affairs involvement beyond basic registrar review. Several registrar policies explicitly require dean review and final approval for non-standard requests or special scheduling exceptions. [4] [3] [1]

Full state table

State	Primary owner	Meaning	Exit paths
Draft	Department scheduler	Section exists but is incomplete and editable	Submit for validation, cancel
Validation in progress	System	Rules engine checks required fields, overlaps, pattern compliance	Validation failed, department ready
Validation failed	Department scheduler	Blocking errors or policy violations found	Return to draft, request exception
Department ready	Department scheduler	Department believes section is ready for unit review	Send to chair review

State	Primary owner	Meaning	Exit paths
Chair review	Department chair/program director	Unit leadership confirms academic need, staffing, and compliance	Approve to dean, return for edits, request exception
Dean review	Dean/associate dean	College-level policy and resource review	Approve to registrar, return for edits, request Academic Affairs exception
Registrar review	Registrar scheduler/admin	Official schedule, policy, and data-quality review	Approve room coordination, return for edits, route to Academic Affairs
Room coordination	Registrar room team / room manager	Central-room assignment and room conflict resolution	Assigned, alternative required, return to registrar
Academic Affairs review	Academic Affairs	Reviews exceptional/non-standard/policy-sensitive requests	Approved to registrar, denied, returned
Approved for publish	Registrar admin	Section is complete, compliant, and publishable	Publish, hold
Published	Registrar admin	Visible in official schedule	Change request, lock
Change requested	Department or registrar	Post-publish modification proposed	Re-validate, registrar review
Locked	Registrar admin	Only emergency/authorized edits allowed	Unlock by registrar admin only
Archived	System/registrar	Historical, read-only	None

Transition rules

```
Draft
  → Validation in progress
  → Cancelled

Validation in progress
  → Validation failed
  → Department ready

Validation failed
  → Draft
  → Exception requested

Department ready
  → Chair review

Chair review
  → Dean review
  → Returned to department
  → Exception requested

Dean review
  → Registrar review
  → Returned to department
  → Academic Affairs review
```

```
Registrar review
  → Room coordination
  → Returned to department
  → Academic Affairs review
  → Approved for publish

Room coordination
  → Approved for publish
  → Returned to registrar
  → Returned to department

Academic Affairs review
  → Registrar review
  → Denied
  → Returned to department

Approved for publish
  → Published
  → On hold

Published
  → Change requested
  → Locked

Change requested
  → Validation in progress
  → Registrar review

Locked
  → Archived
  → Emergency unlock

Denied / Cancelled
  → Archived
```

Exception routing rules

Route to Academic Affairs when:

- A section uses a non-standard meeting pattern during controlled hours.
- A special class type requires nonstandard build treatment.
- Contact-hour or curricular exceptions apply.
- Policy waiver requests exceed dean or registrar authority.

Institutions commonly require standard time blocks for centrally scheduled space and reserve non-standard approvals for registrar or higher academic authority review.[5] [6] [7]

Role-based permission matrix

Role design should distinguish operational creation from formal approval, and separate ordinary edit rights from override and export rights. Education RBAC guidance recommends mapping access by action type and scope rather than broad page access.[8]

Core roles

Role	Typical scope	Function
Department scheduler	Department/program	Builds and submits schedule data
Department chair/program director	Department/program	Academic approval and department oversight
Dean/associate dean	College/school	College-level approval and exception sponsorship
Registrar scheduler	Institution/campus	Policy review and central schedule coordination
Registrar admin	Institution/campus	Final publish authority and official overrides
Academic Affairs reviewer	Institution/college	Reviews special exceptions and policy deviations
Room manager	Central/department room pool	Room assignment and spatial conflict handling
Auditor/read-only reviewer	Scoped	Audit-only access

Permission matrix

Capability	Dept scheduler	Chair	Dean	Registrar scheduler	Registrar admin	Academic Affairs	Room manager
View department sections	Yes	Yes	Yes	Yes	Yes	Scoped	Scoped
View college-wide sections	No	Limited	Yes	Yes	Yes	Scoped	Scoped
Create draft sections	Yes	Limited	No	Yes	Yes	No	No
Edit draft sections	Yes	Limited	No	Yes	Yes	No	No
Submit to workflow	Yes	Yes	Yes	Yes	Yes	No	No
Return to department	No	Yes	Yes	Yes	Yes	Yes	No
Approve to next stage	No	Yes	Yes	Yes	Yes	Yes	No
Request exception	Yes	Yes	Yes	Yes	Yes	Yes	No
Sponsor non-standard request	No	Yes	Yes	Yes	Yes	Yes	No

Capability	Dept scheduler	Chair	Dean	Registrar scheduler	Registrar admin	Academic Affairs	Room manager
Assign department rooms	Scoped	No	No	Yes	Yes	No	Yes
Assign central rooms	No	No	No	Yes	Yes	No	Yes
Override room conflict	No	No	No	Limited	Yes	No	Limited
Override time-pattern rule	No	No	Limited	Limited	Yes	Yes	No
Publish official schedule	No	No	No	No	Yes	No	No
Lock published schedule	No	No	No	Limited	Yes	No	No
Export full schedule dataset	Scoped	Scoped	Scoped	Yes	Yes	Limited	Scoped
View audit logs	No	No	Limited	Limited	Yes	Limited	No

Department scheduler matrix

Department schedulers need a narrower, action-focused matrix.

Function	Allowed	Notes
View prior-term schedules	Yes	Scoped to department/program
Copy prior-term sections	Yes	Creates new draft only
Create/edit draft section	Yes	Before approval or when returned
Change instructor assignment	Yes	Subject to conflict validation
Request central room	Yes	Does not guarantee assignment
Assign department-owned room	Yes	Only within allowed room pool
Submit comments to reviewers	Yes	Workflow-visible notes
Submit full department package	Yes	Can be term-wide or section-level
Edit after chair/dean submission	No	Unless returned or reopened
Override compliance rule	No	Must request exception
Publish or lock	No	Registrar-admin functions
Bulk export beyond scope	No	Scoped exports only

Transition rules in detail

Department stage rules

A section can move from `Draft` to `Validation in progress` only if all required structural fields exist:

- Course and section identity.
- Term and part of term.
- Capacity.
- Meeting pattern or approved placeholder.
- Instructor assignment, if institutionally required at that stage.
- Room request or location status.
- Ownership unit and funding/attribute fields where required.

A section cannot move to `Department ready` if blocking validation errors remain. Scheduling rules are commonly used to prevent time conflicts, double-booked instructors, and policy deviations before formal review. [9]

Chair/dean rules

Chair review should confirm:

- Academic necessity.
- Faculty staffing plausibility.
- Internal program balance.
- Department-level compliance.

Dean review should confirm:

- College-level balance.
- Resource feasibility.
- Exceptional requests worthy of escalation.
- Compliance with peak-time or distribution rules where used.

Some institutions explicitly place department chairs and deans in the scheduling compliance chain and require dean approval for exceptions or non-standard times. [10] [1] [4]

Registrar rules

Registrar review should enforce:

- Approved time-block model.
- Contact-hour compatibility.
- Cross-list and shared-room integrity.

- Data completeness for publication.
- Central classroom eligibility.

Standard meeting-pattern enforcement is a widely used registrar control for minimizing conflicts and improving classroom utilization. [5] [9] [10]

Publish rules

A section should enter `Approved for publish` only if:

- All required approvals are complete.
- All blocking conflicts are resolved or explicitly waived.
- Room assignment is final or intentionally marked as TBA under policy.
- Exception cases have written disposition.
- Publication snapshot can be generated.

Conflict detection engine

The conflict engine should run at save time, submission time, nightly batch scans, and pre-publish certification. Scheduling best practices explicitly call out the need to detect double-booked instructors, room overlaps, and rule violations automatically. [11] [9]

Conflict categories

Conflict type	Severity	Example
Instructor overlap	Blocking	Same instructor assigned to overlapping meetings
Room overlap	Blocking	Two sections in same room at same time
Capacity mismatch	Warning/blocking	Enrollment cap exceeds room capacity
Pattern violation	Blocking or warning	Non-standard time in controlled hours
Contact-hour mismatch	Blocking	Weekly minutes do not satisfy credit/contact expectations
Prime-time imbalance	Warning/escalation	Department exceeds allowed peak-hour share
Room-feature mismatch	Warning/blocking	Requested accessibility/AV feature unavailable
Cross-listed split error	Blocking	Shared sections misaligned in time/room
Travel-time issue	Warning	Back-to-back instructor rooms too far apart

Overlap detection logic

For each section meeting:

1. Expand the meeting into normalized meeting instances using term dates, meeting days, start time, and end time.
2. Group by resource:

- Instructor.
- Room.
- Cross-list group.

3. Detect time interval intersections on shared days within the same active date range.
4. Exclude cancelled or inactive meetings.
5. Apply exception/waiver rules where an approved override exists.
6. Emit blocking or warning conflicts with human-readable explanations.

Pseudocode

```
function overlaps(a, b) {
  return (
    a.day === b.day &&
    a.start < b.end &&
    b.start < a.end &&
    a.dateStart <= b.dateEnd &&
    b.dateStart <= a.dateEnd
  );
}

function detectInstructorConflicts(meetings) {
  const byInstructor = groupBy(
    meetings.filter(m => m.instructorId),
    m => m.instructorId
  );

  const conflicts = [];

  for (const instructorId of Object.keys(byInstructor)) {
    const items = byInstructor[instructorId];
    for (let i = 0; i < items.length; i++) {
      for (let j = i + 1; j < items.length; j++) {
        if (overlaps(items[i], items[j])) {
          conflicts.push({
            type: "INSTRUCTOR_OVERLAP",
            severity: "BLOCKING",
            instructorId,
            leftSectionId: items[i].sectionId,
            rightSectionId: items[j].sectionId
          });
        }
      }
    }
  }

  return conflicts;
}

function detectRoomConflicts(meetings) {
  const byRoom = groupBy(
    meetings.filter(m => m.roomId),
```

```
    m => m.roomId
  );

  const conflicts = [];

  for (const roomId of Object.keys(byRoom)) {
    const items = byRoom[roomId];
    for (let i = 0; i < items.length; i++) {
      for (let j = i + 1; j < items.length; j++) {
        if (overlaps(items[i], items[j])) {
          conflicts.push({
            type: "ROOM_OVERLAP",
            severity: "BLOCKING",
            roomId,
            leftSectionId: items[i].sectionId,
            rightSectionId: items[j].sectionId
          });
        }
      }
    }
  }

  return conflicts;
}
```

Rule engine extensions

Add policy rules for:

- Standard meeting blocks by term and room pool.
- Peak-hour distribution targets.
- Minimum turnaround between room uses.
- Instructor travel constraints between buildings.
- Department-owned room priority.
- Centrally scheduled room priority for standard-pattern sections.

Policies from multiple universities show that standard-time compliance and central-room priority rules are common, and nonstandard sections may lose room priority or need explicit approval.[6] [7] [10]

Operational outputs

The engine should produce:

- Inline form errors during editing.
- A conflict queue for department schedulers.
- A registrar resolution dashboard.
- Pre-publish certification reports.
- Trend analytics on repeated rule violations.

That gives departments immediate feedback while preserving central oversight.

Recommended UI surfaces

Use four key workflow views:

- **Department workspace:** draft, validation, corrections, submit.
- **Approver inbox:** chair, dean, Academic Affairs, registrar decisions.
- **Conflict console:** grouped by instructor, room, time-pattern, and severity.
- **Publication console:** final checks, release preview, and snapshot creation.

This mirrors real institutional workflows where departments submit, approvers review, registrars coordinate rooms and compliance, and final publication happens centrally. [2] [3]

⁂

1. https://www.stmarytx.edu/policies/academic-affairs/class-scheduling-policy/
2. https://registrar.tulane.edu/sites/default/files/2023-11/fall-2024-class-scheduling-timeline.pdf
3. https://it.osu.edu/student-information-system-sis/sis-room-class-scheduling/1-course-class-workflow
4. https://bookstack.clarkson.edu/books/sas-student-achievement-services/page/class-scheduling-guidelines
5. https://registrar.uiowa.edu/faculty-and-staff/classroom-scheduling/course-scheduling-policies-and-procedures
6. https://registrar.arizona.edu/faculty-staff-resources/room-course-scheduling/room-course-scheduling-policies-procedures
7. https://registrar.washu.edu/classroom-scheduling-standards-expectations/
8. https://bluepes.com/blog/role-based-access-education-systems
9. https://coursedog.com/articles/best-practices-for-strategic-scheduling
10. https://www.unomaha.edu/registrar/faculty-and-staff/clss/course-schedule-building-guidelines.php
11. https://moderncampus.com/blog/streamlining-course-scheduling-for-universities-challenges-and-solutions.html
12. https://registrar.vanderbilt.edu/registration/yes-help/Undergrad-Prof-Class-Enrollment-GPCM-Approval-workflow.php